

4	(a) (i)	50 mT	1	
	(ii)	flux linkage = BAN $= 50 \times 10^{-3} \times 0.4 \times 10^{-4} \times 150 = 3.0 \times 10^{-4} \text{ Wb}$	1	
		<i>(allow 49 mT $\rightarrow 2.94 \times 10^{-4} \text{ Wb}$ or 51 mT $\rightarrow 3.06 \times 10^{-4} \text{ Wb}$)</i>	1	[3]
	(b)	e.m.f./induced voltage <i>(do not allow current)</i> proportional/equal to rate of change/cutting of flux (linkage)	1	
			1	[2]
	(c) (i)	new flux linkage = $8.0 \times 10^{-3} \times 0.4 \times 10^{-4} \times 150$ $= 4.8 \times 10^{-5} \text{ Wb}$ change = $2.52 \times 10^{-4} \text{ Wb}$	1	
			1	[2]
	(ii)	e.m.f. = $(2.52 \times 10^{-4})/0.30$ $= 8.4 \times 10^{-4} \text{ V}$	1	
			1	[2]
	(d)	<i>either</i> for a small change in distance x (change in) flux linkage decreases as distance increases so speed must increase to keep rate of change constant	1	
			1	
			1	[3]
	or	(change in) flux linkage decreases as distance increases at constant speed, e.m.f./flux linkage decreases as x increases so increase speed to keep rate constant	(1)	
			(1)	
			(1)	